EMERGENCY_STOP
0: STOP
1: GO

CORD

ON/OFF
0: OFF
1: ON

TEAM

PWM

EYE
Other MCU

LIDAR

BATTERY

OTOS

J103
Conn_01x02

J104
Conn_01x02

J105
Conn_01x02

J106
Conn_01x02

TP101

Conn_01x03
J107

+5V_SW

J108
BM04B-SRSS-TB

+3.3V

I2C_EYE_SCL
I2C_EYE_SDA

+5V

JP102

J109
Conn_01x04

JP101

VBAT_IN

J101
Conn_01x02

J116
BM04B-SRSS-TB

I2C_OTOS_SCL
I2C_OTOS_SDA

J102
USB_C_Receptacle_USB2.0_14P

VBUS

CC1
CC2

D-
D-
D+
D+

SHIELD
GND

SH
A1

5.1kΩ
R101

5.1kΩ
R102

IO

DSI_STOP
DSI_CORD_IN
DSI_ON
DSI_TEAM_IN
PWM_S1_OUT
PWM_S2_OUT

PAMI_STOP
DSI_CORD
PAMI_nON
DSI_TEAM
PWM_S1
PWM_S2
PWM_LED_RGB

File: 0400_IOEXT.kicad_sch

R113
3.3kΩ

R115
3.3kΩ

Power

PAMI_nON
PAMI_STOP

VBAT_MEAS
IPAMI_MEAS

File: 0500_Power.kicad_sch

R114
3.3kΩ

R116
3.3kΩ

MCU

DSI_STOP
DSI_CORD
DSI_ON
DSI_TEAM
PWM_S1
PWM_S2
PWM_LED
I2C_EYE_SCL
I2C_EYE_SDA
LIDAR_UART_TX
VBAT_MEAS
IPAMI_MEAS
I2C_OTOS_SCL
I2C_OTOS_SDA
USB_D-
USB_D+

File: 0100_MCU.kicad_sch

I2C_CAM_SDA
I2C_CAM_SCL

R109
3.3kΩ

R110
3.3kΩ

M1_STEP
M1_DIR
MOT_UART_TX
MOT_UART_RX

R111
1kΩ

R112
0Ω

M2_STEP
M2_DIR

M3_STEP
M3_DIR
M_nEN

R105
100kΩ

R107
100kΩ

R103
100kΩ

R104
100kΩ

R106
100kΩ

R108
100kΩ

I2C_CAM

I2C_CAM_SDA
I2C_CAM_SCL

I2C_CAM_SCL0
I2C_CAM_SDA0
I2C_CAM_SCL1
I2C_CAM_SDA1
I2C_CAM_SCL2
I2C_CAM_SDA2

BM04B-SRSS-TB
J110
CAM1

BM04B-SRSS-TB
J111
CAM2

BM04B-SRSS-TB
J112
CAM3

File: 0300_I2C_CAM.kicad_sch

Motor1

STEPPER_STEP
STEPPER_DIR
UART_RTX
UART_AD0
UART_AD1

STEPPER_1B
STEPPER_1A
STEPPER_2A
STEPPER_2B
DRV_nEN

J113
Conn_01x04
MOTOR1

File: 0200_Motors.kicad_sch

Motor2

J114
Conn_01x04
MOTOR2

File: 0200_Motors.kicad_sch

Motor3

J115
Conn_01x04
MOTOR3

File: 0200_Motors.kicad_sch

PAMI_STOP

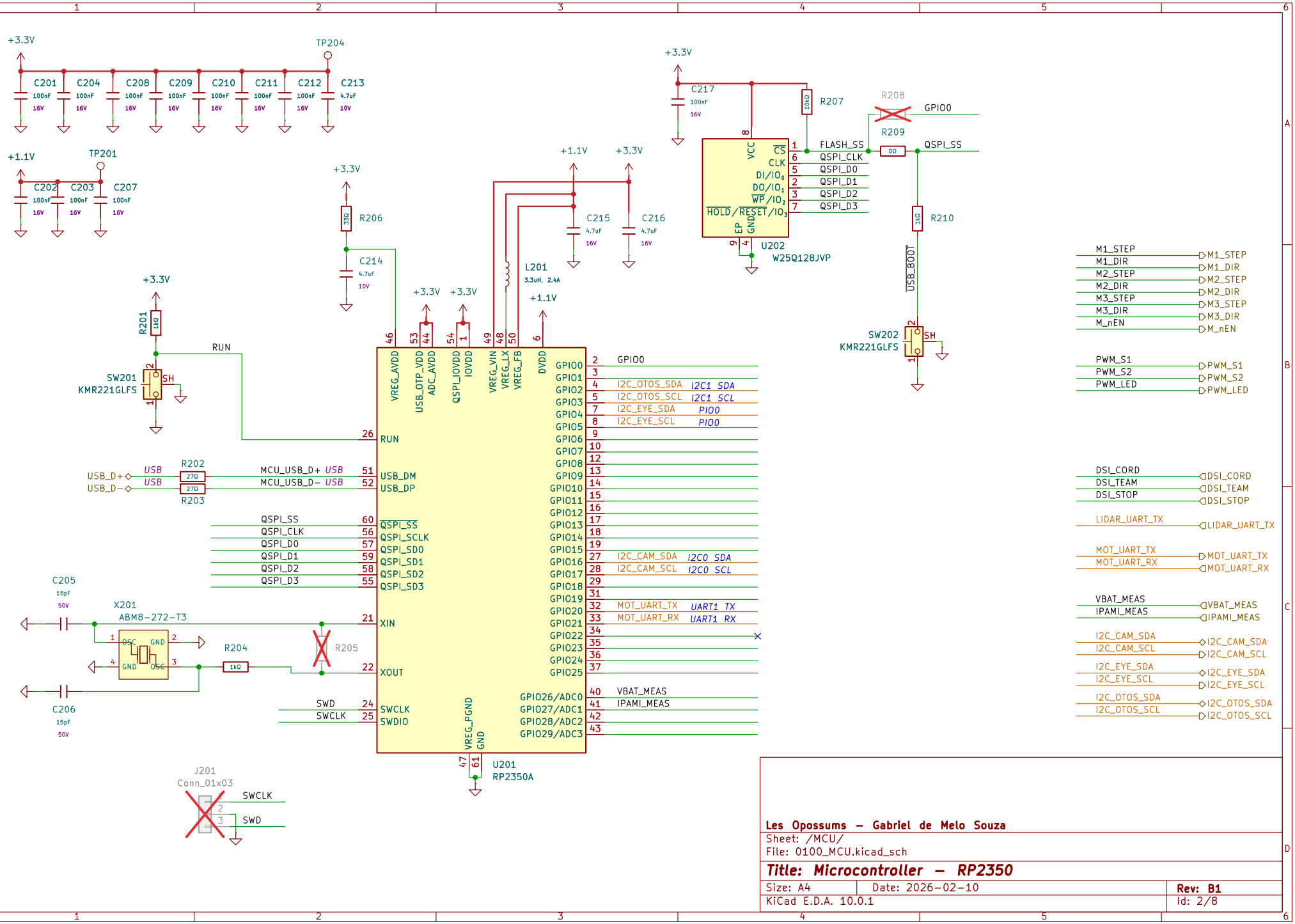
JP103

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Sheet: /
File: PAMI_2027_A1.kicad_sch

Title: PAMI – Main

Size: A3	Date: 2026-02-10	**Rev: B1**
KiCad E.D.A. 10.0.1		Id: 1/8



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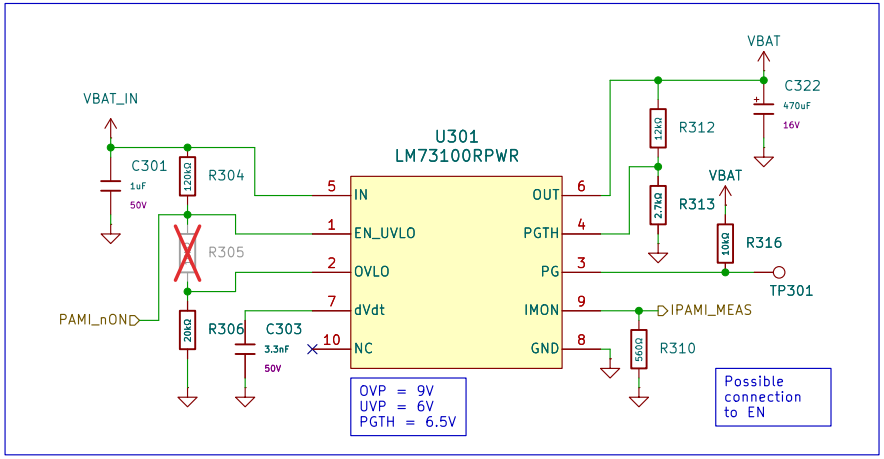
Sheet: /MCU/

File: 0100_MCU.kicad_sch

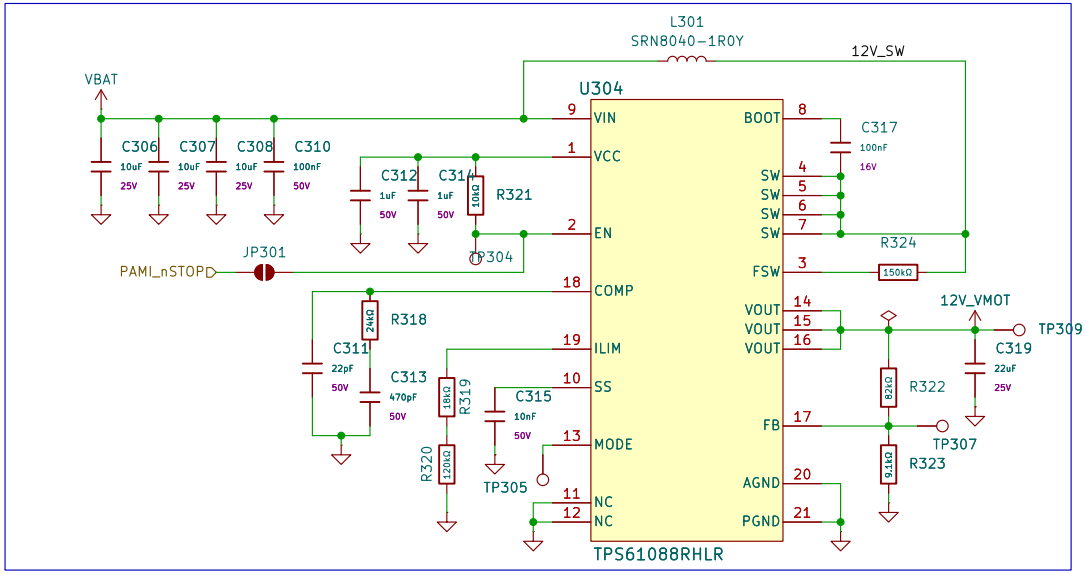
Title: Microcontroller – RP2350

Size: A4	Date: 2026-02-10	**Rev: B1**
KiCad E.D.A. 10.0.1		Id: 2/8

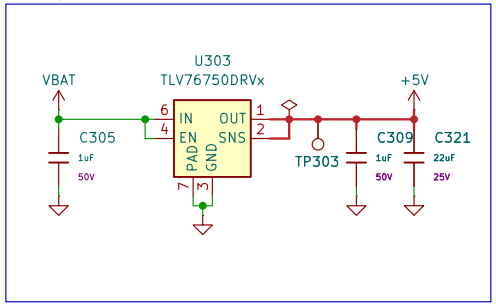
Load switch – VBAT



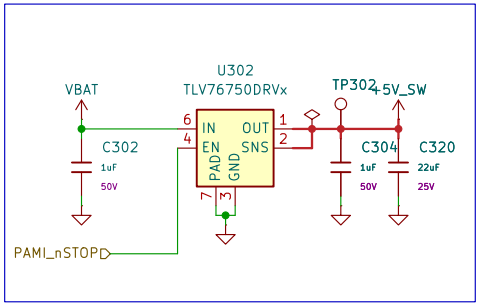
+12V Stepper Motors generation



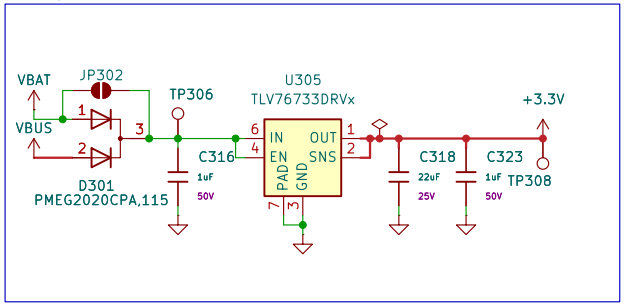
Always ON 5V LDO



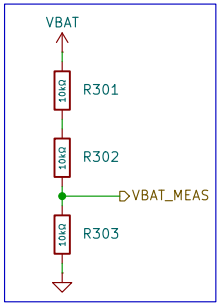
Switched 5V – +5V_SW



Always ON 3.3V LDO



VBAT Measurement



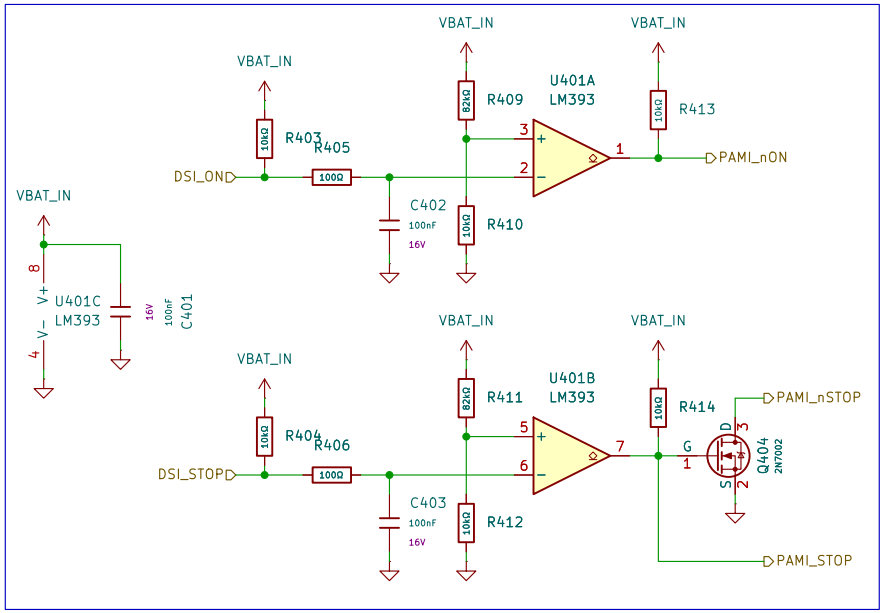
Les Opossums – Gabriel de Melo Souza

Sheet: /Power/
File: 0500_Power.kicad_sch

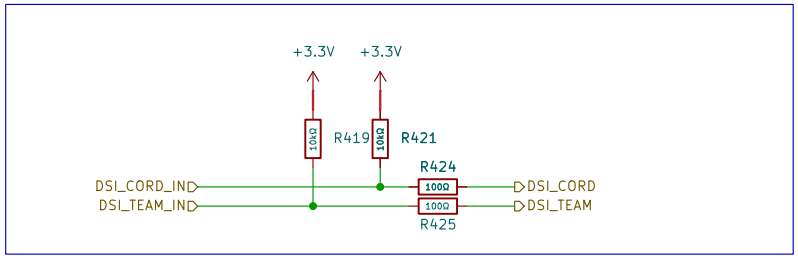
Title: Power

Size: A4	Date: 2026-02-10	**Rev: B1**
KiCad E.D.A. 10.0.1		Id: 3/8

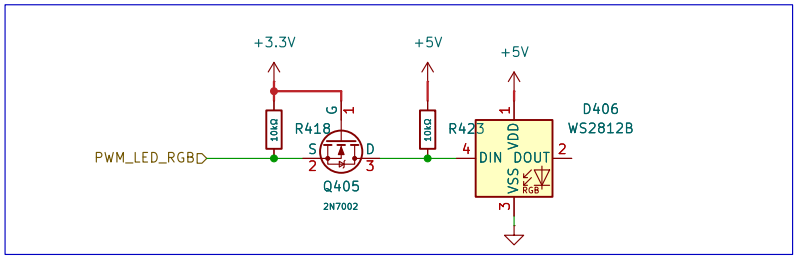
Button inputs



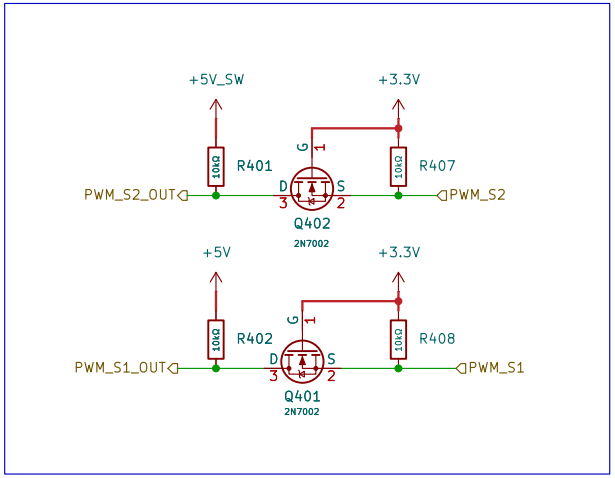
Discrete inputs



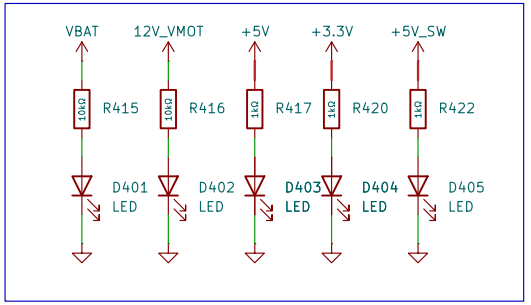
RGB LED



Servo outputs



LED indicators

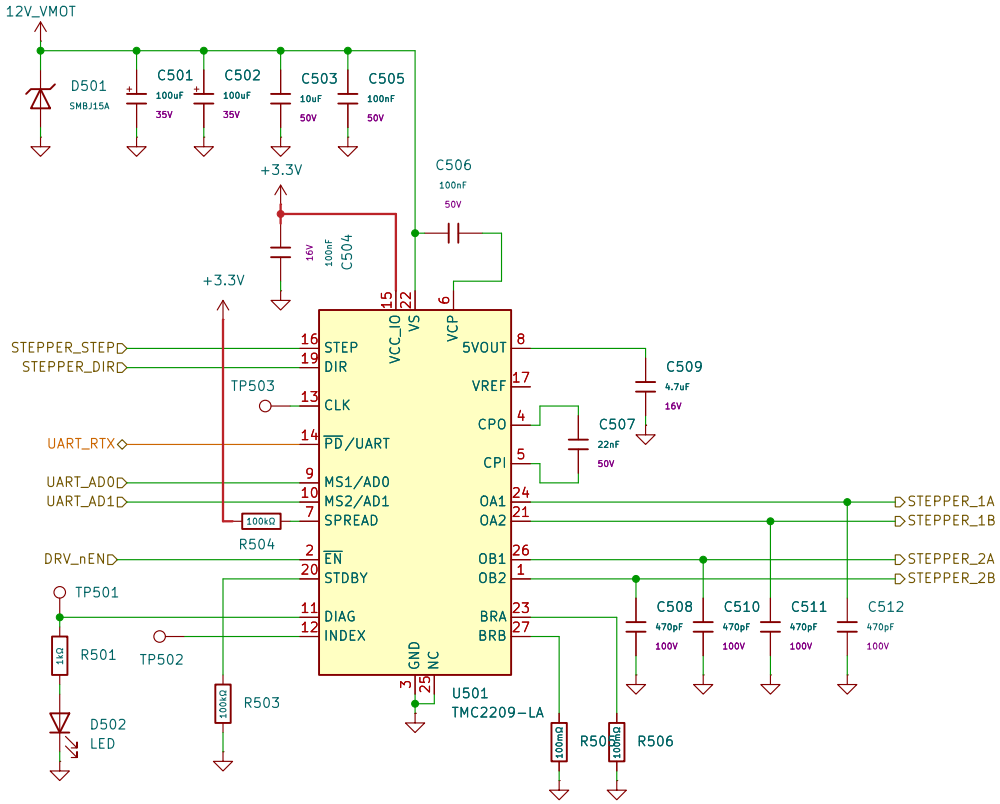


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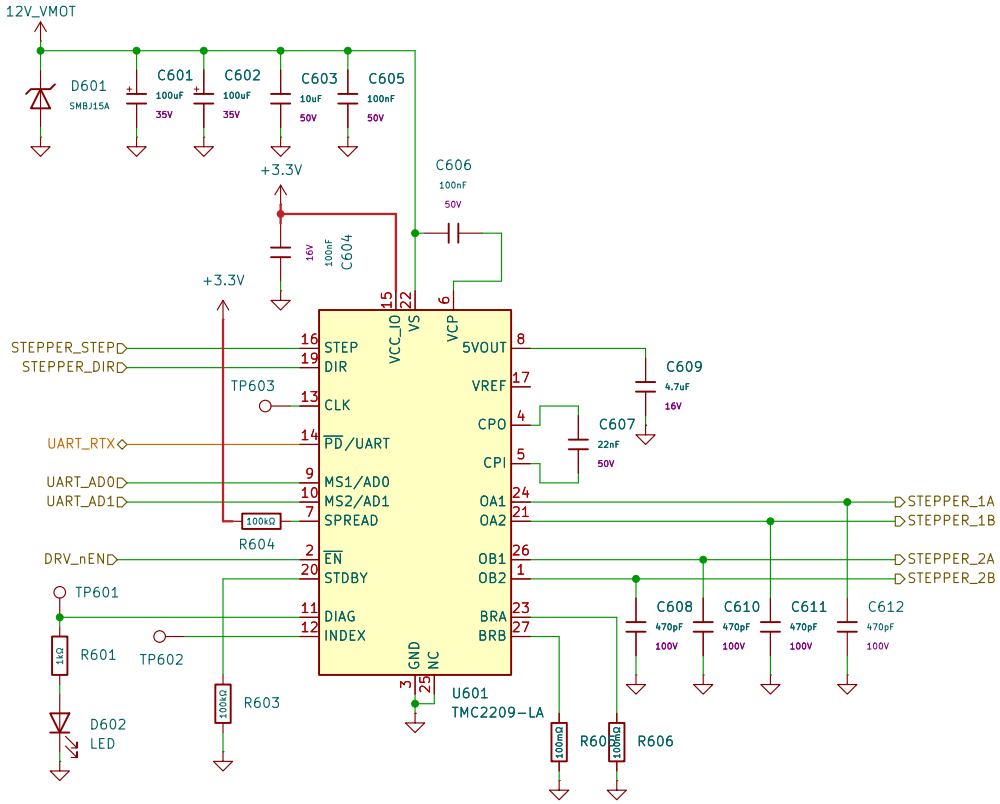
Sheet: /IO/
File: 0400_IOEXT.kicad_sch

Title: IO Adaptation Stage

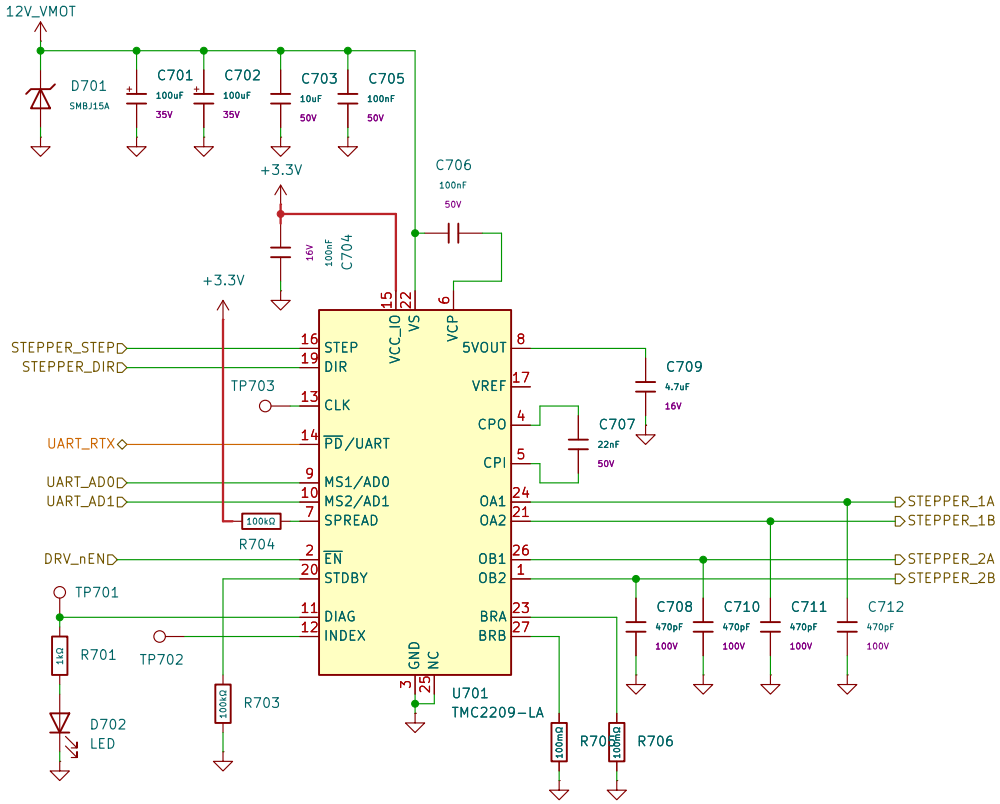
Size: A4	Date: 2026-02-10	**Rev: B1**
KiCad E.D.A. 10.0.1		Id: 4/8



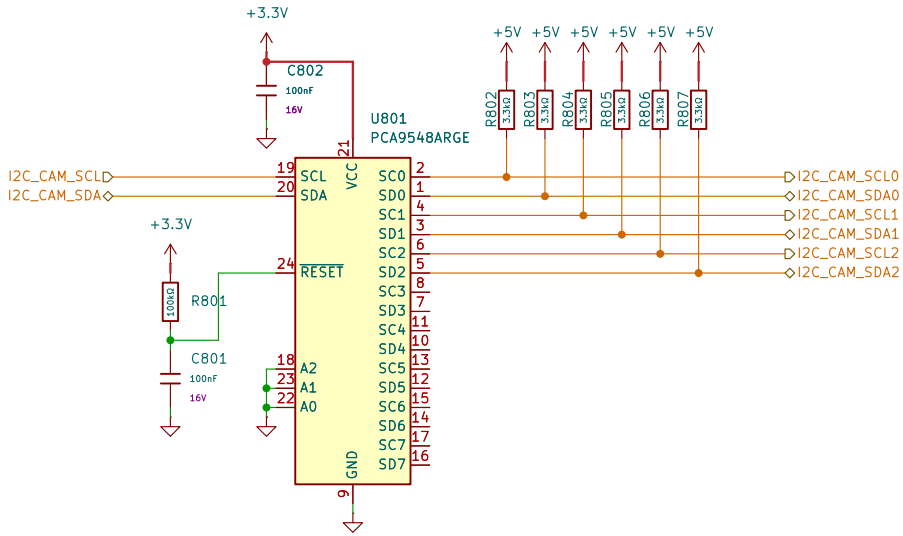
Les Opossums – Gabriel de Melo Souza		
Sheet: /Motor1/		
File: 0200_Motors.kicad_sch		
Title: Stepper Motor		
Size: A4	Date: 2026-02-10	**Rev: B1**
KiCad E.D.A. 10.0.1		Id: 5/8



Les Opossums – Gabriel de Melo Souza	
Sheet: /Motor2/	
File: 0200_Motors.kicad_sch	
Title: Stepper Motor	
Size: A4	Date: 2026-02-10
KiCad E.D.A. 10.0.1	Rev: B1
	Id: 6/8



Les Opossums – Gabriel de Melo Souza		
Sheet: /Motor3/		
File: 0200_Motors.kicad_sch		
Title: Stepper Motor		
Size: A4	Date: 2026-02-10	**Rev: B1**
KiCad E.D.A. 10.0.1		Id: 7/8



Sheet: /I2C_CAM/
File: 0300_I2C_CAM.kicad_sch

Title:

Size: A4	Date: 2026-02-10	**Rev:**
KiCad E.D.A. 10.0.1		Id: 8/8